

Moamen Moustafa Basyoni

C# Engineer | Code Analysis & LLM Evaluation

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PROFESSIONAL SUMMARY

C# engineer whose current work sits at the intersection this role describes: reading unfamiliar code at scale, and judging what LLMs do with it. Built an AI code-documentation system combining Roslyn static analysis and code metrics with LLM interpretation, an automated pull-request review that comments line-by-line and checks implementations against their stated requirements, and a code graph spanning a 38-module application. Runs LLM evaluation and observability in production through LangSmith and LangFuse, with multi-model orchestration across OpenAI and OpenRouter. Three years of C# and .NET behind it, including extensive work extending and refactoring a large, unfamiliar legacy codebase.

PROFESSIONAL EXPERIENCE

R&D Engineer — Code Analysis & AI Tooling · Andalusia Health Business Solution (AHBS)

Dec 2023 – Present · Alexandria, Egypt

Code analysis and LLM evaluation

- Built an **AI code-documentation system** combining **Roslyn** static analysis and code metrics with LLM interpretation — producing documentation carrying both a structural and a semantic view of the codebase.
- Built **automated pull-request review** integrated into Azure DevOps: opens threads, posts overall and **line-level comments**, and checks the implementation against the business requirements in the linked work item. This is, in practice, continuous assessment of code changes against intent.
- Built a **code graph spanning 38 modules** of a production application, mapping code to business capability.
- **LLM evaluation and observability** with LangSmith and LangFuse — prompt telemetry, token usage tracking, and comparison across models; **multi-model orchestration** over OpenAI and OpenRouter with model selection, rate limiting and cost control.
- Built five custom Model Context Protocol (MCP) servers, including a **database server** and a **Playwright** server used for AI-driven test analysis and automation.
- Applies guardrails, prompt-injection mitigation and human-in-the-loop review.

C# / .NET engineering

- Build and refactor services on **C#, ASP.NET Core** and **.NET 8/9/10**, with Entity Framework Core over SQL Server and PostgreSQL.
- Work extensively inside a **large, mature codebase that predated me** — modernized 5 core and 7+ additional modules from .NET Framework 4.5 / ASP.NET Core 2.2, including replacing a legacy MS DTC design with Temporal Saga orchestration. Reading unfamiliar code and forming a correct model of it quickly is the core skill.
- **Debugging and testing**: production issue triage and performance tuning; xUnit and pytest; a runtime rule engine benchmarked to a **60%** improvement with BenchmarkDotNet.
- **Git and Docker** daily — migrated source control from a legacy TFVC monolith to Azure DevOps Git multi-repo with a private NuGet feed, and deliver containerized CI/CD across Dev, Staging and Pre-Live.
- Conduct technical code reviews; mentored 25 engineers across 6 teams.

Technical Partner (equity) — Trastain

Feb 2025 – Aug 2025 · Cairo, Egypt (Remote) · Part-time

- Built a booking platform from scratch, including RAG-based matching and multimodal AI search.
- **Remote collaboration** across time zones with a distributed founding team.

Software Engineer — MYM (Make Your Miracle)

Jun 2023 – Oct 2023 · Giza, Egypt

- Backend systems for e-commerce and booking platforms, delivered end to end.

TECHNICAL SKILLS

- **Languages**: C#, Python, SQL, TypeScript, JavaScript
- **C# / .NET**: ASP.NET Core, .NET 8/9/10, .NET Framework 4.5, Entity Framework Core, **Roslyn** (static analysis and metrics), xUnit, BenchmarkDotNet
- **Code analysis**: Roslyn-based static analysis, code graphs, automated PR review, implementation-versus-requirement checking, debugging and production triage
- **LLM evaluation & tooling**: LangSmith, LangFuse, prompt telemetry, multi-model orchestration (OpenAI, OpenRouter), model selection, Model Context Protocol (MCP), LangGraph, CrewAI, RAG, guardrails
- **AI-assisted development**: Claude Code (daily), GitHub Copilot, Codex, OpenCode
- **Tooling**: **Git**, GitHub, GitHub Actions, Azure DevOps, **Docker**, TFS/TFVC migration
- **Testing**: xUnit, pytest, pytest-asyncio, Playwright (via MCP-driven automation)

EDUCATION

B.Sc. Computer Engineering — Behera Higher Institute (BHI), Alexandria, Egypt · 2023 Grade: B+ · GPA 3.2 / 4.0

PUBLICATION & LANGUAGES

- *IoT-Enabled E-Prescription Management and Dispensing Machine Monitoring* — IEEE, 2024
- Arabic (Native) · English (B2, Upper Intermediate)